Isha Kulkarni

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Summary

AI Engineer with 2 years of expertise using Python to design, build, train, evaluate, deploy, and maintain machine learning, deep learning, and generative AI systems: LLM powered applications, retrieval augmented generation (RAG) pipelines, and autonomous AI agents shipped as cloud native microservices on AWS and Azure. Strong background in artificial intelligence: LLM integration, LangChain orchestration, multi agent systems, prompt engineering, and vector embeddings for enterprise stakeholders.

Technical Skills

Programming Languages: Python • JavaScript (React/Node.js) • TypeScript • Java
LLM & GenAI: LangChain • RAG • Agentic AI • Prompt Engineering • Tool Calling
ML & NLP: TensorFlow • PyTorch • Scikit-learn • NLP • Computer Vision • Transformers
MLOps & Evaluation: LLM Evaluation • Benchmarking • Model Monitoring • Observability
Vector Databases: ChromaDB • MongoDB • MySQL • Hybrid Retrieval • Semantic Chunking
Cloud & Production: AWS • Azure • FastAPI • Docker • Microservices • CI/CD • REST APIs

Experience

Persistent Systems Limited — Aug 2023 – July 2025
Software Engineer — AI and Machine Learning (GenAI) — *Pune, India*

- Designed multi agent generative AI systems with LangChain tool calling, custom agent memory, and intelligent task routing to automate end to end AI task execution, reducing manual intervention by 70 percent at production scale.
- Built cloud native, LLM powered Python pipelines integrating OpenAI and Hugging Face foundation models via LangChain orchestration, automating QA workflows across 3 product lines and reducing manual QA workload by 60 percent.
- Shipped 5+ production RAG retrieval pipelines with semantic chunking, hybrid retrieval, and vector embeddings on AWS and Azure, improving document search relevance by 45 percent across enterprise client deployments.
- Deployed LLM evaluation harnesses with automated scoring, hallucination detection, benchmarking, and experiment tracking using custom metrics, improving model accuracy by 50 percent across 10+ deployed language model applications.
- Architected asynchronous FastAPI microservices for model inference on AWS EC2 and Azure App Services, achieving 99.9 percent uptime with under 500ms response latency at sustained production load.
- Led 50+ client facing workshops translating artificial intelligence use cases into production roadmaps, driving GenAI adoption from PoC to deployment with product and engineering stakeholders in agile teams across 10+ enterprise accounts.

Revyamp Consultancy — Nov 2022 – July 2023
Software Engineer Intern — *Pune, India*

- Engineered a Python OCR invoice processing pipeline using Tesseract and OpenCV computer vision to extract structured data from unstructured documents, cutting manual data entry time by 60 percent while processing 500+ invoices daily.
- Built a React and Node.js monitoring dashboard for real time invoice pipeline observability, enabling finance teams to track processing status and reducing query resolution time by 40 percent.

Projects

Crawl AI — Website to RAG Knowledge System | *Python • LangChain • FastAPI • ChromaDB* | GitHub — 2026

- Built a Python web crawler with recursive URL discovery and deduplication to index 100+ website pages into a ChromaDB vector database with semantic chunking and embedding generation, cutting knowledge base search time by 40 percent.
- Deployed a FastAPI RAG service using LangChain retrieval chains and context aware prompt templates, achieving 95 percent answer accuracy for domain specific customer queries over large unstructured knowledge bases.

Test Script Migrator — LLM Test Migration Tool | *Python, FastAPI • Streamlit* | GitHub — 2025

- Built a Python LLM framework using prompt chaining and AST aware parsing to autonomously convert legacy Protractor test suites to modern Selenium, handling 90 percent of migration patterns without any manual intervention.
- Deployed an asynchronous FastAPI backend and Streamlit frontend for fully on premise local LLM inference, enabling secure enterprise usage with zero data egress and under 3s latency per migrated test file end to end.
- Designed a modular pipeline separating parsing, LLM inference, and code generation into independent stages with automated testing and validation, reducing migration errors by 75 percent across 500+ test scripts.

Education

University of Southampton — Sep 2025 – Present
Master of Science in Computer Science — *Southampton, United Kingdom*

- Completed coursework in Machine Learning, NLP, Deep Learning, and Advanced Databases, training and evaluating neural networks and transformers in agentic AI research.

Savitribai Phule Pune University — Aug 2019 – Jul 2023
Bachelor of Technology in Information Technology — *Pune, India*

- Achieved 9.02 / 10.0 CGPA, ranked in the top 3 percentile of the cohort consecutively across all semesters.